

PRODUCT REQUIREMENTS DOCUMENT

AI-Powered Interview Prep

for Working Professionals

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1. Executive Summary

Working professionals preparing for software engineering interviews face a unique challenge: they understand core CS theory from their day jobs but lack structured problem-solving practice and have severely limited time. Existing platforms (Programming Pathshala, Striver, Scaler, AlgoExpert) offer static curricula designed for students with 6–10 hours daily — a poor fit for someone juggling a full-time job, commute, and personal responsibilities.

This product is an AI-powered interview preparation companion that assesses a professional's current skill level, sets a personalized goal-driven plan, and continuously adapts the learning path based on performance, available time, and progress. It integrates directly with the user's calendar for session blocking, sends timely reminders, tracks progress toward deadlines, and provides motivational nudges to maintain consistency.

The core differentiator is intelligent personalization: the app does not just hand you a syllabus — it thinks for you, deciding what to study, when, and in what order, recalibrating every time you complete a session or miss one.

2. Problem Statement

2.1 Target User

Software engineers with 1–10 years of experience currently employed full-time, who want to switch companies or aim for FAANG/top-tier roles. They typically have 1–2 hours per day (and slightly more on weekends) for preparation.

2.2 Pain Points

- No structured roadmap: They know they need to practice DSA, system design, etc., but don't know where to start or what order to follow given their existing knowledge.
- Time scarcity: Fixed, lengthy course curricula feel overwhelming. They need a plan that fits their actual schedule, not a generic 12-week bootcamp.
- Inconsistency: Without external accountability, most professionals start strong but lose momentum within 2–3 weeks.
- No feedback loop: They solve problems but have no way to know if they're improving, stagnating, or focusing on the wrong areas.
- Calendar friction: Preparation sessions don't get blocked on the calendar, so they get displaced by work meetings and personal tasks.

3. Product Vision

"Imagine a mentor who knows exactly where you stand, what you need to work on next, fits your schedule perfectly, and never lets you slack off — that's what this app is."

The product sits at the intersection of adaptive learning AI, calendar management, and behavioral motivation — a category we call “Intelligent Prep Orchestration.”

4. Suggested Product Names

Based on the product’s identity and positioning, here are five name candidates:

Name	Rationale
PrepFlow	Suggests a flowing, adaptive preparation journey
CodeStride	Steady progress in coding — stride by stride
InterviewPilot	AI as your co-pilot for interview preparation
Preplytic	Prep + Analytic — data-driven preparation
CodeCraft Pro	Crafting coding skills for professionals

Recommendation: PrepFlow or CodeStride — both are short, memorable, and convey the adaptive nature of the product.

5. User Personas

5.1 Persona A — “The Busy Switcher”

- Name: Rahul, 27, SDE-1 at a mid-size product company in Bangalore
- Goal: Switch to a FAANG company within 4 months
- Available time: 1.5 hours on weekdays, 3 hours on weekends
- Strengths: Solid understanding of arrays, strings, basic trees from daily work
- Weakness: Never practiced graph algorithms, dynamic programming, or system design formally
- Frustration: Bought an online course but it starts from arrays — 80% of the content is stuff he already knows

5.2 Persona B — “The Experienced Returner”

- Name: Priya, 32, Senior Engineer at a services company in Pune
- Goal: Crack a product company interview after a 5-year gap from competitive coding
- Available time: 1 hour on weekdays only (has a toddler, weekends are family time)
- Strengths: Strong system design intuition, good at LLD from work
- Weakness: Rusty on DSA, never practiced on LeetCode-style problems
- Frustration: Feels overwhelmed by the sheer volume of problems on LeetCode; doesn’t know which 100 problems would give her the best ROI

6. Core Features

6.1 Intelligent Onboarding & Assessment

The onboarding flow is the foundation of the entire product. It must feel lightweight (under 15 minutes) while extracting enough signal to build a meaningful initial plan.

Step 1: Goal Setting

- Target role: SDE-1, SDE-2, SDE-3, Staff, or “General interview readiness”
- Target companies: FAANG, unicorn startups, specific company names
- Timeline: Target interview date or flexible (“as soon as ready”)
- Prep areas: DSA, System Design (HLD), Low-Level Design (LLD), CS Fundamentals, Frontend/Backend specific, Behavioral — user selects which apply

Step 2: Skill Assessment

An adaptive diagnostic quiz (10–15 questions) that adjusts difficulty based on responses. The AI maps the user to a proficiency matrix across topics:

Topic Area	Sub-Topics	Proficiency Levels
DSA	Arrays, Trees, Graphs, DP, Greedy, Backtracking, etc.	Beginner / Intermediate / Advanced / Expert
System Design	Scalability, Databases, Caching, Load Balancing, etc.	Beginner / Intermediate / Advanced / Expert
LLD / OOP	Design Patterns, SOLID, Class Design, etc.	Beginner / Intermediate / Advanced / Expert
CS Fundamentals	OS, DBMS, Networking, Concurrency	Beginner / Intermediate / Advanced / Expert

Step 3: Schedule Input

- Connect Google Calendar (or manually input available hours per day/week)
- Set preferred study times (morning, evening, late night)
- Indicate hard constraints (e.g., “no Sundays”, “only after 9 PM”)

6.2 AI-Generated Adaptive Study Plan

This is the core intelligence layer. Based on the assessment, goals, and schedule, the AI generates a week-by-week plan with daily sessions.

Plan Generation Logic

- Skip mastered topics: If the user scored Advanced on Arrays, the plan starts them on harder array problems or skips to weaker areas entirely

- Prioritize high-ROI topics: For a FAANG SDE-2 target, DP and Graphs get higher weight than basic string manipulation
- Interleave topics: Uses spaced repetition principles — revisit weaker topics periodically rather than front-loading everything
- Respect time constraints: If a user has only 1 hour/day, each session is exactly 1 hour with 1–2 problems and a short concept review
- Adapt continuously: After each session, the AI re-evaluates. If the user struggled with a DP problem, tomorrow's plan may include an easier DP warm-up before moving on

Session Structure

Each daily session is structured as: a 5-minute concept recap (short notes or a 2-min video), followed by 1–3 curated problems (difficulty-appropriate), then a self-assessment prompt (“How confident do you feel about this topic now?” which feeds back into the AI). The total duration is calibrated to the user's available window.

6.3 Progress Tracker & Dashboard

- Visual skill map showing proficiency across all topics (radar chart), updating in real time as the user completes sessions
- Streak counter and consistency score (similar to GitHub contribution heatmap)
- Weekly progress summary: problems solved, topics covered, estimated readiness percentage
- Milestone markers: “You're now interview-ready for Arrays & Strings at FAANG level”
- Deadline countdown: days remaining to target interview date with a pacing indicator (on track / behind / ahead)

6.4 Calendar Integration & Smart Scheduling

This is a critical differentiator. The app doesn't just suggest a plan — it blocks time on the user's real calendar.

- Google Calendar and Outlook integration via OAuth
- Auto-block study sessions as calendar events with topic details in the description
- Intelligent rescheduling: if a work meeting gets added over a study slot, the AI finds the next available window and moves the session automatically
- Weekend catch-up: if the user missed sessions during the week, the AI proposes a compressed weekend plan
- Pre-session notification (15 min before) with the topic and problem links

6.5 Reminders & Motivational Engine

Consistency is the number one predictor of interview success. The motivation system is designed around behavioral psychology:

- Smart push notifications: contextual reminders that adapt to the user's response patterns (if they ignore evening reminders, try morning ones)
- Streak protection: "You've studied 12 days in a row — don't break it! Today's session is just 30 minutes."
- Peer benchmarking (opt-in): "Professionals with your profile typically solve 3 problems/day at this stage"
- Milestone celebrations: confetti, badges, and shareable cards when the user hits key milestones
- Gentle nudges after inactivity: escalating reminders — Day 1: encouraging, Day 3: concerned, Day 7: re-engagement offer with a reduced plan

7. User Journey

The end-to-end flow from download to interview readiness:

Phase	User Action	System Response
1	Downloads app and signs up	Welcome screen, asks for role and goal
2	Sets target (e.g., Google SDE-2 in 3 months)	Calculates total preparation effort needed
3	Takes adaptive assessment (10–15 min)	Maps proficiency across all topic areas
4	Inputs schedule / connects calendar	Generates personalized week-by-week study plan
5	Receives first calendar-blocked session	Pre-session notification with topic and problem links
6	Completes session, self-rates confidence	AI adjusts next session difficulty and topic focus
7	Misses a session on Wednesday	AI reschedules to Thursday evening or weekend catch-up
8	Checks dashboard after 2 weeks	Sees skill map improvement, streak count, pacing status
9	Hits “Interview Ready” milestone for DSA	Celebration, shifts focus to System Design
10	Reaches target date	Mock interview recommendation and final readiness report

8. Competitive Landscape

The table below maps existing players against the core capabilities this product uniquely combines:

Feature	This Product	LeetCode	Prog. Pathshala	StudyFetch	Reclaim.ai
AI Skill Assessment	✓	✗	✗	✗	✗
Adaptive Study Plan	✓	✗	✗	✓	✗
Calendar Blocking	✓	✗	✗	Partial	✓
Smart Rescheduling	✓	✗	✗	✗	✓
Progress Tracking	✓	Basic	Basic	✓	✗
Motivational Engine	✓	✗	✗	✗	✗

Interview-Specific	✓	✓	✓	✗	✗
For Working Pros	✓	✗	✗	✗	✓

No single competitor combines all of these capabilities. LeetCode has the problems but no planning. Reclaim has the calendar but no interview context. Programming Pathshala has the curriculum but no AI personalization. This product brings all three together.

9. Technical Architecture (High Level)

9.1 Platform

Mobile-first (React Native for iOS and Android) with a companion web dashboard (Next.js). The mobile app is the primary interface for daily sessions, notifications, and quick progress checks. The web dashboard is for detailed analytics, plan customization, and calendar management.

9.2 Core Services

- **Assessment Engine:** Adaptive testing using Item Response Theory (IRT) to estimate proficiency with minimal questions
- **Plan Generator:** Constraint-based scheduling AI that takes proficiency gaps, time availability, goal deadline, and topic dependencies as inputs to produce an optimal study sequence
- **Adaptation Service:** After each session, recalculates topic weights and difficulty levels using a lightweight ML model trained on completion data and self-assessment signals
- **Calendar Sync Service:** OAuth-based integration with Google Calendar and Microsoft Outlook; handles event creation, conflict detection, and rescheduling
- **Notification Engine:** Multi-channel (push, email, SMS) with send-time optimization based on user engagement patterns
- **Content Service:** Manages curated problem bank, concept notes, and video links; supports tagging by topic, difficulty, and company frequency

9.3 AI/ML Stack

- LLM layer (GPT-4 / Claude API) for natural language goal setting, plan explanation, and motivational messages
- Custom recommendation model for problem selection (collaborative filtering + content-based, similar to how Spotify recommends songs)
- Spaced repetition algorithm (modified SM-2) for topic revisit scheduling

10. Success Metrics

Metric	Target (Month 3)	Target (Month 12)
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Weekly Active Users (WAU)	5,000	50,000
7-Day Retention	40%	55%
Average Sessions / Week	4.5	5.5
Plan Completion Rate	25%	45%
NPS Score	45+	60+
Calendar Integration Adoption	30% of users	60% of users
Paid Conversion (Freemium)	8%	15%

11. Monetization Strategy

11.1 Freemium Model

The free tier includes the initial assessment, a basic study plan (first 2 weeks), limited daily problems (2/day), and basic progress tracking. This gives users enough value to experience the product's intelligence before hitting the paywall.

11.2 Premium Tier (₹499–999/month or ₹3,999–5,999/year)

- Full adaptive study plan with unlimited duration
- Calendar integration and smart rescheduling
- Unlimited daily problems with detailed AI explanations
- Advanced analytics dashboard (topic-level heatmaps, time-spent analysis, predicted readiness)
- Motivational engine with streak protection and peer benchmarking
- Mock interview scheduling and feedback

11.3 Revenue Projections

Assuming 50,000 WAU at month 12 with 12% paid conversion at an average revenue per user (ARPU) of ₹600/month, monthly recurring revenue would reach approximately ₹36 lakhs (~\$43K). With year-over-year growth driven by word-of-mouth and content marketing, ARR could reach ₹5–8 crore by year 2.

12. Risks & Mitigations

Risk	Impact	Mitigation
Cold start: AI plan feels generic before enough data	High	Use onboarding assessment aggressively; seed with expert-curated default paths per role
User drops off after initial	High	Motivational engine, streak

enthusiasm		mechanics, gentle re-engagement nudges, weekend catch-up plans
Calendar permission anxiety (privacy)	Medium	Offer manual schedule input as alternative; clear data usage policy; minimal calendar access scope
Content curation quality at scale	Medium	Start with a curated bank of 500–800 high-quality problems; expand gradually with community contributions and quality reviews
LLM API costs eat into margins	Medium	Cache common responses; use smaller models for routine tasks; reserve GPT-4/Claude for complex interactions

13. MVP Scope (Phase 1 — 8 Weeks)

The minimum viable product focuses on proving the core value proposition: that an AI-generated, time-aware study plan significantly outperforms self-directed preparation.

In Scope

Onboarding flow with goal setting and adaptive assessment (DSA-focused initially). AI-generated weekly study plan with daily sessions. Curated problem bank (300+ problems tagged by topic, difficulty, and company). Progress dashboard with skill map and streak tracking. Google Calendar integration for session blocking. Push notification reminders. Basic motivational nudges (streak alerts, milestone celebrations).

Out of Scope (Phase 2+)

System Design and LLD modules. Mock interview scheduling. Peer benchmarking. Microsoft Outlook integration. Community features. Advanced analytics. Behavioral interview prep.

14. Indicative Timeline

Week	Milestone	Owner
1–2	Finalize assessment question bank; design onboarding UI	Product + Content
3–4	Build assessment engine and plan generator backend	Engineering
5–6	Build mobile app (onboarding, dashboard, session view)	Engineering
7	Google Calendar integration; notification system	Engineering

8	Internal testing, bug fixes, content QA	All
9	Closed beta with 50–100 working professionals	Product + Growth
10–12	Iterate on feedback, public launch	All

15. Open Questions

1. Should the problem bank be proprietary or integrate with LeetCode/Codeforces via their APIs? (Trade-off: control vs. familiarity)
2. Should we include an in-app code editor, or link out to LeetCode/IDE for problem solving? (Trade-off: seamless experience vs. build complexity)
3. What is the right balance between AI automation and user control over the study plan? (Some users may want to override AI suggestions)
4. Should the motivational engine use social features (leaderboards, study groups) or stay individual-focused for privacy?
5. Pricing sensitivity for the Indian market — is ₹499/month the right entry point, or should we go lower to maximize adoption?

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